Time series: Exercises on ARMA properties

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Lecture notes 10

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The homework is provided on the last slide

Representation of ARMA as MA

Task. Check that the ARMA process $x_t = 0.5x_{t-1} + \epsilon_t - 0.3\epsilon_{t-1}$ is causal and represent it as an MA process.

For the given process, we have $a(z) = 1 - 0.5z$. The root is $z_1 = 2$, which is located outside the unit circle, i.e. $|z_1| > 1$. Thus, the process x_t is causal.

Since $(1 - 0.5B)x_t = (1 - 0.3B)\epsilon_t$ and $\frac{1}{1-u} = 1 + u + u^2 + u^3 + \ldots$, we have

Covariance function of ARMA via MA form

Task. Compute the covariance function of the ARMA process $x_t = 0.5x_{t-1} + \epsilon_t - 0.3\epsilon_{t-1}$ via the MA representation.

In slide 2, we obtained that x_t has the form $x_t = \sum_{j=0}^{\infty} c_j \epsilon_{t-j}$, where $c_0 = 1$ and $c_j = 0.2 \cdot 0.5^{j-1}$ for $j = 1, 2, \ldots$.

Then we have

$$
\begin{aligned}
R(k) &= \sigma^2 \sum_{s=0}^{\infty} c_s c_{s+|k|} \\
&= \sigma^2 [0.2 \cdot 0.5^{|k|-1} + \sum_{s=1}^{\infty} 0.2 \cdot 0.5^{s-1} 0.2 \cdot 0.5^{s+|k|-1}] \\
&= \\
&=
\end{aligned}
$$

Representation of ARMA as AR

Task. Check that the ARMA process $x_t = 0.5x_{t-1} + \epsilon_t - 0.3\epsilon_{t-1}$ is invertible and represent it as an AR process.

For the given process, we have $b(z) = 1 - 0.3z$. The root is $z_1 = 10/3$, which is located outside the unit circle, i.e. $|z_1| > 1$. Thus, this process is invertible.

Since $(1 - 0.5B)x_t = (1 - 0.3B)\epsilon_t$ and $\frac{1}{1-u} = 1 + u + u^2 + u^3 + \ldots$, we have

Checking causality and invertibility, ARMA(1,1)

Task. Identify if an ARMA(1,1) process $x_t - 1.5x_{t-1} = \epsilon_t + 0.5\epsilon_{t-1}$ is causal or not. Why?

For this process, the polynomial has the root $z_1 =$, which is located inside the unit circle, i.e. $|z_1| < 1$. Thus, the process x_t is .

Task. Identify if an ARMA(1,1) process $x_t - 0.3x_{t-1} = \epsilon_t + 1.8\epsilon_{t-1}$ is invertible or not. Why?

For this process, the polynomial has the root $z_1 =$, which is located inside the unit circle, i.e. $|z_1| < 1$. Thus, the process x_t is .

Variance and correlation function of AR(2)

Task. Find the variance and the correlation function of the AR(2) process $x_t = \alpha(x_{t-1} - x_{t-2}) + \epsilon_t$.

The 1-st and 2-nd Yule-Walker equations are

$$\rho(1) = \alpha - \alpha\rho(1),$$
$$\rho(2) = \alpha\rho(1) - \alpha.$$

Covariance function and spectrum of MA(2)

Task. Find the covariance function and the spectral density of the MA(2) process $x_t = \epsilon_t + \beta\epsilon_{t-1} + \gamma\epsilon_{t-2}$.

Direct calculation gives

$$R(0) = \mathbf{E}x_t x_t = \mathbf{E}(\epsilon_t + \beta\epsilon_{t-1} + \gamma\epsilon_{t-2})(\epsilon_t + \beta\epsilon_{t-1} + \gamma\epsilon_{t-2})$$

$=$

$$R(1) = \mathbf{E}x_t x_{t-1} = \mathbf{E}(\epsilon_t + \beta\epsilon_{t-1} + \gamma\epsilon_{t-2})(\epsilon_{t-1} + \beta\epsilon_{t-2} + \gamma\epsilon_{t-3})$$

$=$

$$R(2) = \mathbf{E}x_t x_{t-2} = \mathbf{E}(\epsilon_t + \beta\epsilon_{t-1} + \gamma\epsilon_{t-2})(\epsilon_{t-2} + \beta\epsilon_{t-3} + \gamma\epsilon_{t-4})$$

$=$

Then $R(k) = 0$ for $k = 3, 4, \ldots$.

The spectral density is $f(\lambda) = \frac{1}{2\pi}\sum_{k=-\infty}^{\infty} R(k)\cos(\lambda k)$

$=$

Covariance function for given spectrum

Task. Let x_t be a weakly stationary process with discrete time and spectral density $f(\lambda) = \pi - |\lambda|$. Find the covariance function of this process.

As given on slide 18 of Lecture 3, the covariance function is computed by

$$R(k) = \int_{-\pi}^{\pi} \cos(k\lambda) f(\lambda) d\lambda.$$

Causality of AR(2) with given coefficients

Task. Identify the range for α when the AR(2) process $x_t = \alpha x_{t-1} + \alpha^2 x_{t-2} + \epsilon_t$ is causal.

The polynomial $a(z) = 1 - a_1 z - a_2 z^2$ with $a_1 = \alpha$ and $a_2 = \alpha^2$ has the roots $z_1 = \frac{a_1 + \sqrt{a_1^2 + 4a_2}}{-2a_2}$, $z_2 = \frac{a_1 - \sqrt{a_1^2 + 4a_2}}{-2a_2}$.

R(0) and R(1) for ARMA(1,1)

Task. Find two values of the covariance function $R(0)$ and $R(1)$ for the ARMA(1,1) process $x_t = \alpha x_{t-1} + \epsilon_t + \beta\epsilon_{t-1}$.

First, we find
$R(0) = \mathbf{E}x_t x_t = \mathbf{E}(\alpha x_{t-1} + \epsilon_t + \beta\epsilon_{t-1})(\alpha x_{t-1} + \epsilon_t + \beta\epsilon_{t-1})$.
Expanding the brackets and using independence, we obtain

$$\begin{aligned} R(0) &= \\ &= \end{aligned}$$

Thus, $R(0) = \sigma^2 \frac{1+\beta^2+2\alpha\beta}{1-\alpha^2}$.
Multiplying the given equation by x_{t-1} and taking $\mathbf{E}$, we get

$$\mathbf{E}x_t x_{t-1} \quad =$$

that is,
Thus, $R(1) = \sigma^2(\alpha \frac{1+\beta^2+2\alpha\beta}{1-\alpha^2} + \beta)$.

MA representation of AR(2)

Task. Compute the coefficients c_0, c_1, c_2, c_3 of the MA representation of the AR(2) process $x_t - x_{t-1} + 0.5x_{t-2} = \epsilon_t$.

Since the AR equation can be written as $(1-B+0.5B^2)x_t = \epsilon_t$ and $\frac{1}{1-u} = 1 + u + u^2 + u^3 + \ldots$, we have

$$\begin{aligned} x_t &= \frac{1}{1-[B-0.5B^2]}\epsilon_t \\ &= \end{aligned}$$

Since we are interested in terms with B^j, $j \leq 3$, we have

$$\begin{aligned} x_t &= \\ &= \end{aligned}$$

Thus, $c_0 = 1,\ c_1 = 1,\ c_2 = 0.5,\ c_3 = 0.$

Covariance function of AR(3)

Task. Find the covariance function of the AR(3) process $x_t = \alpha x_{t-3} + \epsilon_t$.

Since the given equation can be written as $(1 - \alpha B^3)x_t = \epsilon_t$, we derive the following MA representation

$$x_t = \frac{1}{1 - \alpha B^3}\epsilon_t =$$

where $c_0 = 1$, $c_3 = \alpha$, $c_6 = \alpha^2$, $c_{3s} = \alpha^s$ and $c_j = 0$ for $j \notin \{0, 3, 6, 9, \ldots\}$. Since $R(k) = \sigma^2 \sum_{j=0}^{\infty} c_j c_{j+|k|}$, we have

$$R(k) = \begin{cases} r_{k/3}, & |k| \in \{0, 3, 6, 9, \ldots\} \\ 0, & |k| \notin \{0, 3, 6, 9, \ldots\} \end{cases}$$

where $r_s = \sigma^2 \sum_{j=0}^{\infty} \alpha^j \alpha^{j+|s|} =$.

MA Representation of an AR(p) Process

Consider the AR(p) model

$$x_t = a_1 x_{t-1} + a_2 x_{t-2} + \cdots + a_p x_{t-p} + \varepsilon_t,$$

with white noise ε_t. Its MA(∞) representation is

$$x_t = \sum_{j=0}^{\infty} c_j \varepsilon_{t-j},$$

where the coefficients $\{\psi_j\}$ satisfy:

- **Initialization:**

$$c_0 = 1.$$

- **Recursion for $j \geq 1$:**

$$c_j = a_1 c_{j-1} + a_2 c_{j-2} + \cdots + a_p c_{j-p},$$

with $c_k = 0$ for $k < 0$.

MA representation of AR(2)

Task. Compute the coefficients c_0, c_1, c_2, c_3 of the MA representation of the AR(2) process $x_t - x_{t-1} + 0.5x_{t-2} = \epsilon_t$.

Initialization: $c_0 = 1$.

Recursion: $c_j = a_1 c_{j-1} + a_2 c_{j-2}$, $j = 1, 2, \ldots,$ with $c_{-1} = 0$.

$a_1 = 1,\ a_2 = -0.5.$

$c_1 =$

$c_2 =$

$c_3 =$

Thus, $c_0 = 1,\ c_1 = 1,\ c_2 = 0.5,\ c_3 = 0.$

Homework: Learning with AI

Utilize AI tools to master the techniques of representation and covariance analysis for complex ARMA models.

- Provide AI with a specific ARMA(2,1) equation (e.g., $x_t - 0.8x_{t-1} + 0.15x_{t-2} = \epsilon_t + 0.3\epsilon_{t-1}$). Ask it to show the step-by-step expansion of the operator $(1 - 0.8B + 0.15B^2)^{-1}$ to find the first five c_j coefficients.
- Ask AI to derive the analytical formula for $R(0)$ and $R(1)$ for the ARMA(1,1) process. Compare the AI's derivation with the results shown on Slide 10.
- Use AI to generate R-code that plots the roots of the characteristic polynomial $a(z)$ for the AR(2) task on Slide 11. Verify visually if they lie outside the unit circle.
- Ask AI to explain the relationship between the MA(∞) representation and the spectral density function. Request a plot showing how the spectrum of an AR(3) process [Slide 12] changes as α approaches the causality boundary.